

ROBOTIC ARM USING ARDUINO

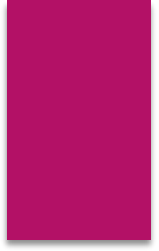
INTRODUCTION:

- ▶ A robotic arm is a programmable mechanical system designed to replicate the functionalities of a human arm. It consists of joints, links, and end-effectors that work together to perform various tasks.
- ▶ They are employed in tasks such as assembly, welding, painting, packaging, and material handling.
- ▶ Robotic arms come in various sizes, ranging from small, lightweight arms suitable for delicate tasks to large, heavy-duty arms capable of lifting and manipulating heavy loads.



WORKING PRINCIPLE:

- ▶ Servo Motors: Each joint of the robotic arm is actuated by a servo motor. Servo motors are commonly used due to their precise control over angular position
- ▶ 1. *Base Rotation (Degree of Freedom 1):* The first servo motor controls the rotation of the robotic arm's base. By rotating the base, the arm can orient itself in different directions around a fixed point.
- ▶ 2. *Shoulder Movement (Degree of Freedom 2):* The second servo motor controls the movement of the arm's shoulder joint. This allows the arm to move up and down or tilt from side to side.
- ▶ 3. *Elbow Movement (Degree of Freedom 3):* The third servo motor controls the elbow joint of the arm. This joint enables the arm to extend or retract, adjusting its reach.
- ▶ 4. *End Effector Rotation or Gripper (Degree of Freedom 4):* The fourth servo motor controls either the rotation of the end effector (the tool or attachment at the end of the arm) or the opening and closing of a gripper. This allows the arm to manipulate objects or perform specific tasks.



PWM Control: Pulse Width Modulation (PWM) is a method used to control the position of servo motors. By varying the width of the pulse, you can set the desired angle of rotation for each servo motor.

Arduino: The Arduino microcontroller is used to generate PWM signals to control the servo motors. It reads input signals, processes them, and outputs corresponding PWM signals to control the servo motors.

Kinematics: The kinematics of the robotic arm determine how each joint's rotation affects the position and orientation of the end effector (the part of the arm that performs tasks). This involves understanding the relationship between joint angles and the position of the end effector in 3D space.

Programming: You need to write a program for the Arduino that interprets desired end effector positions and calculates the required joint angles to achieve those positions. This program will then convert these angles into PWM signals for each servo motor.

CALCULATIONS:

Length of gripper = 6.9 cm (DL)

Length of wrist to elbow = 7.7 cm (CD)

Length of shoulder to elbow = 8 cm (BC)

Length of Base to shoulder = 4.5 cm (AB)

Weight of Motor = 9 gm

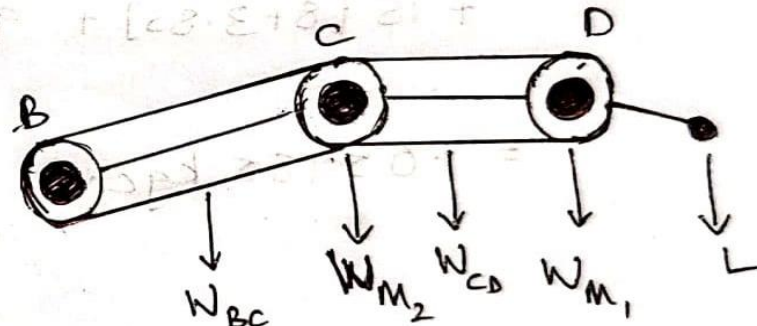
Weight of AB = 17 gm

Weight of BC = 25 gm

Weight of CD = 15 gm

Weight of gripper = 12 gm.

Load = 12 + 3 = 15 gm.



CALCULATIONS:

Calculation of Torque

$$\begin{aligned}T_D &= (L_4 \times L) + \left[\frac{1}{2} \times L_4 \times W_4 \right] \\&= (6.9)(15) + \frac{1}{2}(6.9)(12) \\&= 144.9 \text{ gcm} \\&= 0.1449 \text{ kgcm}\end{aligned}$$

$$\begin{aligned}T_C &= L \times [L_{CD} \times L_4] + W_4 [L_{CD} \times (L_4/2)] + W_M \times L_{CD} \\&\quad + W_{CD} \times (L_{CD}/2)\end{aligned}$$

$$\begin{aligned}&= (15)[7.7 + 6.9] + 12[7.7 + 3.45] + 9 \times 7.7 + 15 \times \left(\frac{7.7}{2}\right) \\&= 0.47985 \text{ kgcm}\end{aligned}$$

CALCULATIONS:

$$T_B = L \times (L_{BC} + L_{CD} + L_A) + W_A \times (L_{BC} + L_{CD} + (L_A/2)) +$$

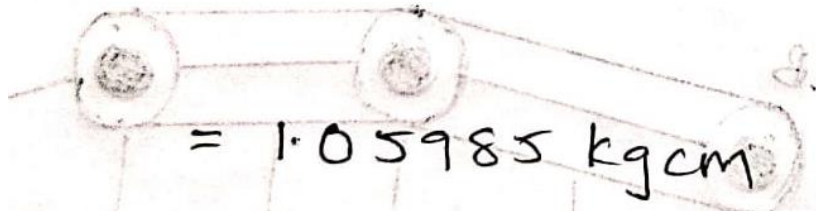
$$W_{M1} \times (L_{BC} + L_{CD}) + W_{CD} \times [L_{BC} + (L_{CD}/2)] +$$

$$W_{M2} \times L_{BC} + W_{BC} \times (L_{BC}/2)$$

$$= (15)(8+7.7+6.9) + 12[8+7.7+3.45] + 9[8+7.7]$$

$$+ 15[8+3.85] + 9 \times 8 + 25[8/2]$$

$$= 1.05985 \text{ kgcm}$$



PWM SERVO MOTOR:

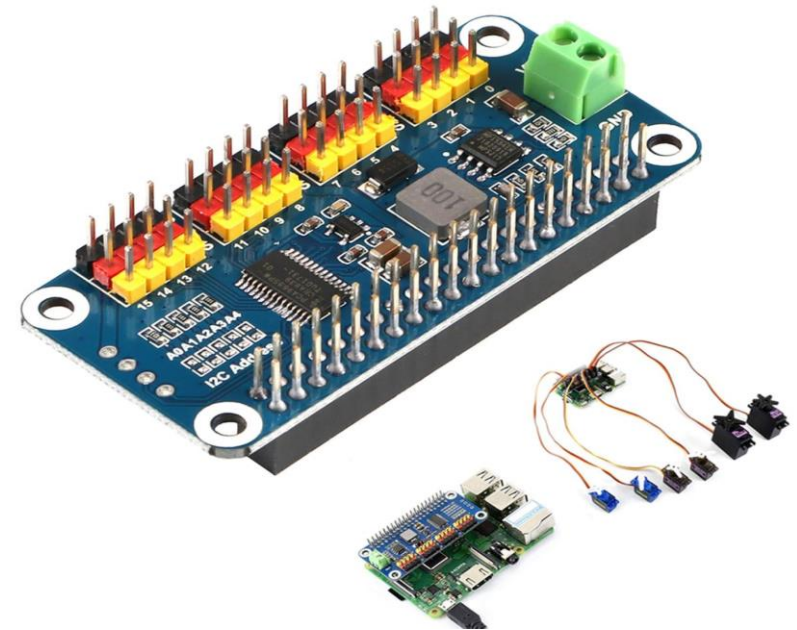
SPECIFICATIONS:

Voltage(VCC): 3.3-5V

Voltage(V+) : 4.8-6VDC

Current : 6mA

Current(PWM controller only) : 400mA



ARDUINO UNO:

► SPECIFICATIONS:

Microcontroller : ATmega328

Operating Voltage : 5V

Recommended Input Voltage : 7-12V

Input Voltage (Limits) : 6-20V

Analog Input Pins : 6

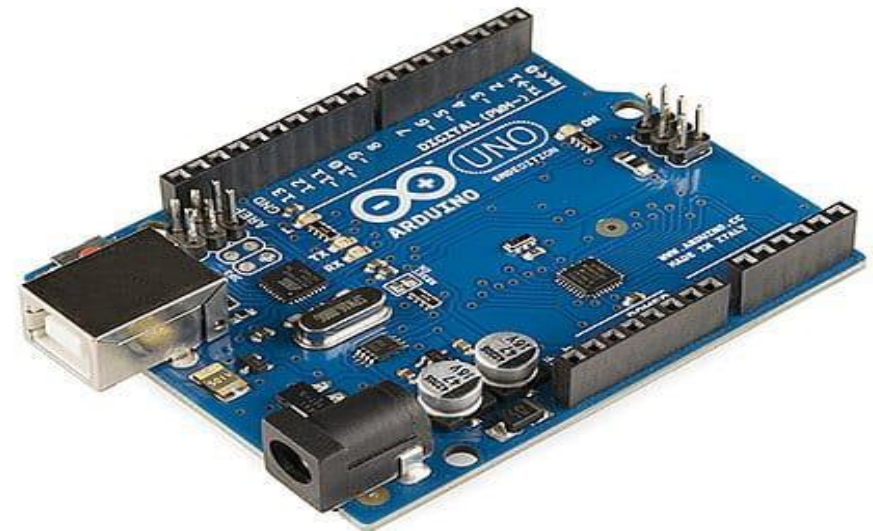
Digital I/O Pin : 14 (Out of which 6 provide PWM output)

DC Current on I/O Pin : 40 mA

DC Current on 3.3V Pin : 50 mA

Flash Memory : 32 KB (0.5 KB is used for Bootloader)

SRAM : 2 KB



SERVOMOTOR:

SPECIFICATIONS:

Weight: 9g.

Size : 22 x 11.5 x 27 mm.

Operating Speed (4.8V no load): 0.12sec/60 degrees.

Stall Torque (4.8V): 17.5oz/in (1.2 kg/cm)

Temperature Range: -30 to +60 Degree C.

Dead Band Width: 7usec.

Operating Voltage: 3.0-7.2 Volts.



POWER SUPPLY:

SPECIFICATIONS:

input Voltage: AC 100 - 264V 50 / 60Hz

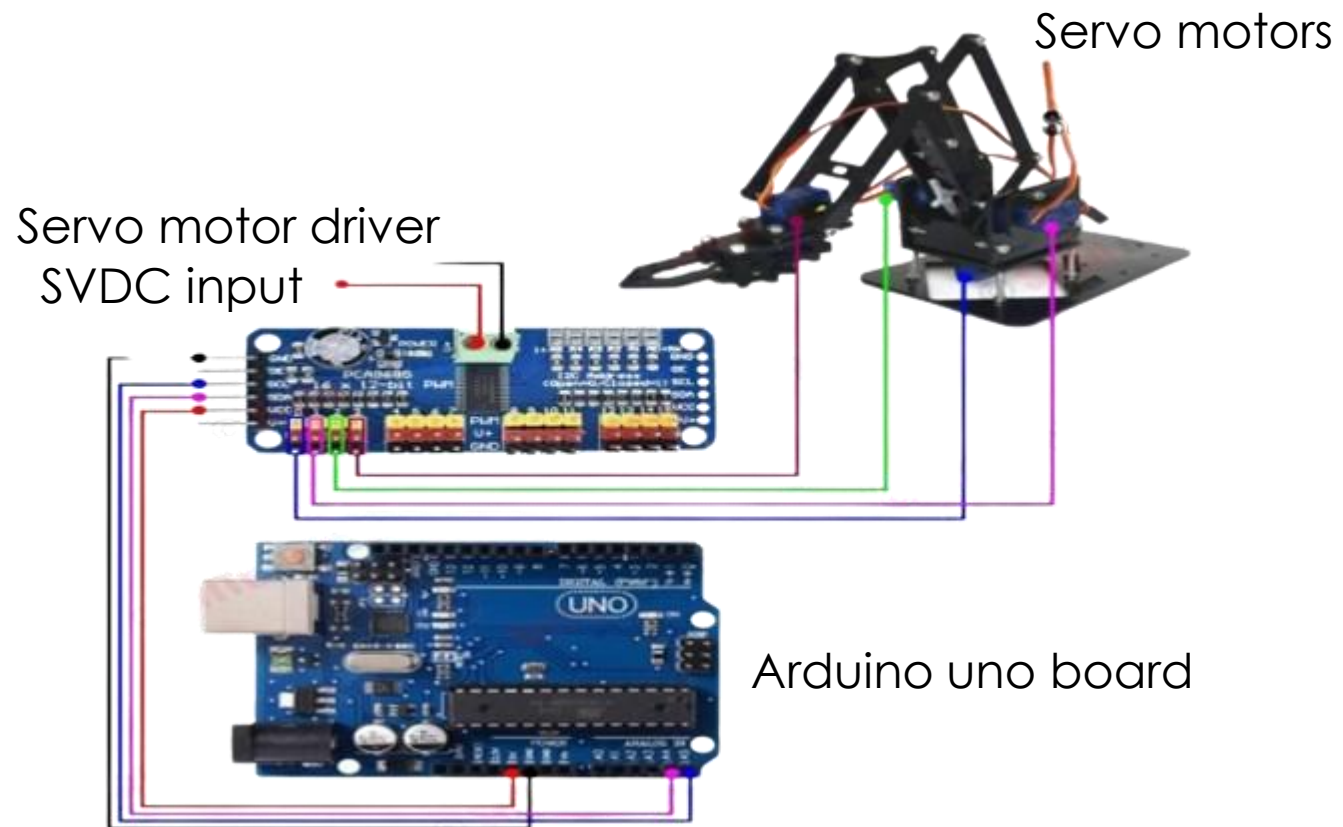
Output Voltage: 5V DC, 5A

Output Current: 5 amps (or 5000 milliamps).

Output Power: 25 watts



CIRCUIT DIAGRAM:



ADVANTAGES:

- ▶ Efficiency: They can work continuously without fatigue, leading to increased productivity and reduced cycle times.
- ▶ Safety: By handling hazardous materials or operating in dangerous environments, robotic arms enhance workplace safety by minimizing the risk to human operators
- ▶ Flexibility: Robotic arms can be reprogrammed or reconfigured to perform different tasks, making them adaptable to changing production needs.
- ▶ Cost-effectiveness: While initial investment costs may be high, robotic arms often provide long-term cost savings through improved efficiency, reduced waste, and lower labor costs.
- ▶ Increased Output: Robotic arms can work faster and for longer durations than human workers, leading to higher output levels and greater throughput.

DISADVANTAGES:

- ▶ Limited Reach: A four-DOF robotic arm may have constraints in reaching certain positions or objects within its workspace due to its restricted range of motion along different axes.
- ▶ Limited Manipulation: With fewer degrees of freedom, the robotic arm may struggle to manipulate objects in complex ways, such as grasping items from various angles or performing intricate assembly tasks.
- ▶ Difficulty in Avoiding Obstacles: With fewer joints and degrees of freedom, it may be more challenging for the robotic arm to navigate around obstacles or obstacles within its workspace, potentially leading to collisions or inefficiencies.
- ▶ Reduced Flexibility: The arm's ability to adapt to different tasks or environments may be limited compared to robotic arms with higher degrees of freedom, which can hinder its versatility in certain applications.

FUTURE ADVANCEMENTS:

- ▶ robotic arm can also be controlled using a Bluetooth controller. This involves connecting the Bluetooth controller to the robotic arm's control system, allowing users to wirelessly send commands to control its movements and actions. This method provides more flexibility and convenience in controlling the robotic arm, especially in applications where direct physical interaction with the arm is not feasible or practical.
- ▶ We can also control a robotic arm using a potentiometer controller by connecting potentiometers to the joints of the arm. As the potentiometers are turned, they generate varying voltage signals corresponding to the position of each joint. These voltage signals are then processed by a microcontroller or a control system to adjust the position and movement of the robotic arm accordingly. This method provides a manual control interface, allowing users to directly manipulate the arm's movements by adjusting the potentiometers.

CONCLUSION:

- ▶ the development of a 4 degrees of freedom (DOF) automatic robotic arm utilizing four servo motors controlled by PWM signals through an Arduino microcontroller, supplied with a 5V DC power source, represents a significant advancement in automation technology. This robotic arm offers versatility, precision, and efficiency in various applications, including industrial automation, assembly lines, and research laboratories. By integrating servo motors controlled by PWM signals, the robotic arm achieves precise and dynamic movements along multiple axes. The Arduino microcontroller facilitates seamless coordination and synchronization of servo motor actions, ensuring smooth and accurate operation. Moreover, the inclusion of an automating code in Arduino enhances the robotic arm's functionality by enabling autonomous operation and predefined task execution. This automation capability reduces the need for manual intervention, increases productivity, and enhances workflow efficiency. Overall, the 4 DOF automatic robotic arm with 4 servo motors and PWM control, powered by a 5V DC power supply and automated through Arduino programming, represents a sophisticated solution for modern automation needs, offering reliability, flexibility, and performance.

REFERENCES:

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- Siciliano, B., & Khatib, O. (Eds.). (2008). Springer Handbook of Robotics. Springer.
- “Arduino Robotics Projects” by Richard Grimmer and “Arduino Project Handbook: 25 Practical Projects to Get You Started”
- “Introduction to Autonomous Robots: Mechanisms, Sensors, Actuators, and Algorithms” by Nikolaus Correll, Bradley Hayes, and Adam To



THANK YOU